The Impact of Temperature on Large Language Models

AI Research Paper Explainer

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In Exploring the impact of temperature on large language models: Hot or cold? Li et al. (2025) examines how adjusting temperature settings influences the behaviour and performance of large language models. LLMs are widely used for tasks such as machine translation, summarization, and question answering, and tuning their hyperparameters can improve performance for specific applications. Temperature is one of the most commonly adjusted hyperparameters, as it directly affects how deterministic or creative a model's responses can be.

In simple terms, the temperature parameter controls how random an LLM's output can be. Higher temperatures produce more varied and creative responses, while lower temperatures make the model more deterministic and predictable. LLMs generate text using next-word prediction, selecting the next token in a sequence based on the probabilities learned during training. For example, given the sequence "We like to go swimming in a ___," the model would assign high probabilities to tokens such as lake, pool, or river, and much lower probabilities to words like car, road, or computer. At low temperatures, the model is more likely to choose the highest-probability tokens, but as temperature increases, the chances of selecting lower-probability tokens also rise.

The research paper also examines different methods for tuning token probabilities, such as Top-K sampling, which selects the next token from the K most likely candidates, and Top-P sampling, which chooses from the smallest set of tokens whose cumulative probability exceeds P. To generate these probabilities, models first produce logits. Logits are values derived from the neural network's calculations. Because logits do not fall within the standard probability range of 0 to 1, they must be converted into probabilities using the Softmax function.

Softmax Function

$$P_i = \frac{e^{z_i}}{\sum_{j=1}^{V} e^{z_j}}$$

Where P_i is the probability of the i-th token.

Where z_i is the value of the logit after the neural network calculations.

Where V is the total number of tokens in the vocabulary.

Softmax Function with Temperature

$$P_i = \frac{e^{z_i/T}}{\sum_{j=1}^{V} e^{z_j/T}}$$

When temperature is applied, each logit is divided by T. Lower temperature values make the probability distribution sharper and more concentrated around the highest-probability tokens, while higher temperatures produce a flatter distribution, increasing the likelihood of selecting a wider range of tokens.

Methodology

This paper aims to determine which temperature settings are best suited for different tasks performed by large language models (LLMs). The authors evaluate six categories of tasks: Causal Reasoning (CR), Creativity (CT), In-Context Learning (ICL), Instruction Following (IF), Machine Translation (MT), and Summarization (SUMM). A table below provides clear definitions and examples for each task.

Task Categories and Descriptions		
Task	Short Explanation	Example
Causal Reasoning	The ability to use the information provided and decide what would logically happen next; understand cause-and-effect relationships.	Recognizing that adding fuel makes a fire grow and predicting what happens when wood is added.
Creativity	The ability to produce new or original ideas that are coherent and meaningful.	Writing an original short story that is entertaining and logically structured.

In-Context Learning	The ability to use information provided in a prompt or passage to answer questions or solve problems.	Reading a story and answering questions based on its content.
Instruction Following	The ability to complete a task exactly as requested.	Instructing an LLM to write a story that is exactly three paragraphs, includes two main characters, and takes place during a snowstorm.
Machine Translation	Using an LLM to translate a text from one language to another.	For example asking the LLM to translate a text from English into Spanish or Chinese.
Summarization	The ability to condense long texts into shorter passages that preserve key information.	Turning a long article into a short paragraph that includes only the key point and main ideas.

Each task is associated with a specific benchmark and dataset used to generate input prompts. The researchers evaluated model performance at different temperatures using LLM-based scoring with carefully designed prompts, exact-match scoring, or established machine translation metrics, depending on the task. They compared output quality across temperature settings ranging from 0.1 to 1.9, increasing in increments of 0.3. The study includes small, medium, and large models from families such as Llama, Mistral, and Qwen, as listed on page 4 of the paper (Li et al., 2025, p. 4).

Takeaways

Results

Instruction Following, Machine Translation, and Summarization showed a strong sensitivity to temperature, with performance varying noticeably across different temperature settings. The results also show that larger models' performance is less affected by changes in temperature, and these models consistently achieve higher performance across all six task categories. For Causal Reasoning, In-Context Learning, and Machine Translation, outcomes varied depending on the model used, suggesting that model selection plays an important role in task performance.

Task-Specific Temperature Findings

Creativity: The optimal temperature for creativity tasks was 1.3, with medium and large models performing best. Increasing the temperature initially improves creativity, but performance begins to decline once the temperature becomes too high.

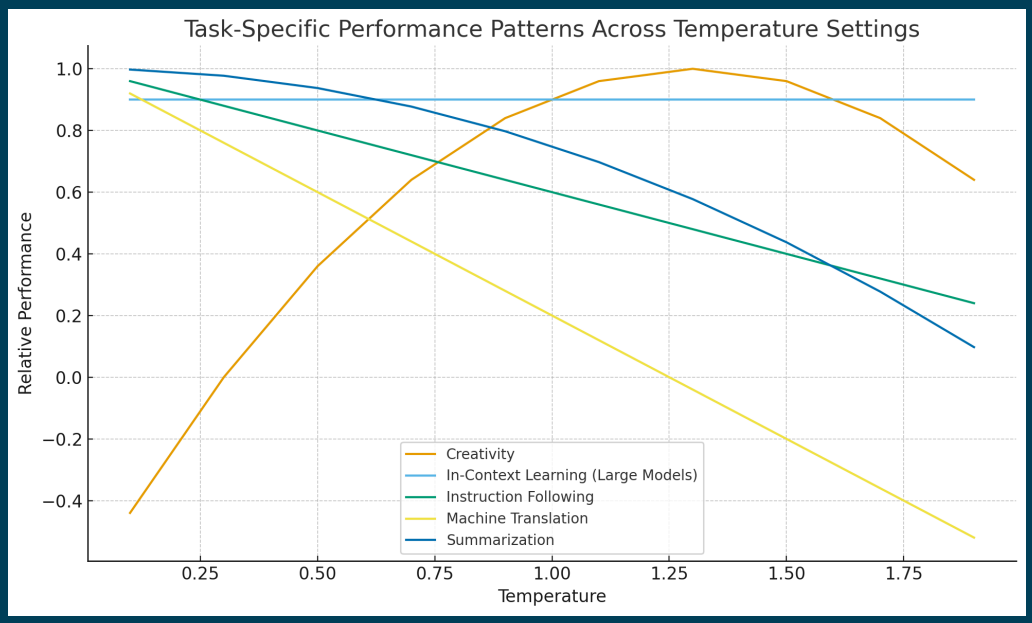
In-Context Learning: Large models achieved the strongest performance and remained stable across different temperatures. Smaller models, however, showed performance degradation as temperature increased.

Instruction Following: For instruction-following tasks, a temperature setting below 1 is recommended to maintain accuracy and reliability.

Machine Translation: Higher temperatures harm machine translation performance. The optimal temperature is close to 0, though results may vary by language depending on the availability of training data.

Summarization: Summarization performance remains relatively stable at lower temperatures but declines sharply as temperature increases. Statistical analysis shows a strong negative correlation between temperature and performance.

Visualization of Findings



Why this Matters at Work

This article advances our understanding of how hyperparameter tuning, specifically temperature, affects large language models (LLMs). Because LLMs function as black-box models, meaning their internal decision-making processes are not directly interpretable, experimenting with temperature provides valuable insight into how to use them more effectively. As more people automate or semi-automate tasks using LLMs, this information can help them optimize performance. For example, if you rely on LLMs for machine translation, lowering the temperature can improve accuracy. The study also reinforces that there is no universal "best" temperature; the ideal setting depends on the task.

Creative fields can also benefit from thoughtful temperature adjustments. The default temperature in tools like ChatGPT and Gemini is 0.7, but this paper suggests that higher temperatures may support more originality in areas such as marketing, design, or creative writing. Raising the temperature increases the likelihood of more varied or unconventional token choices. Conversely, tasks that require precision or consistency, such as translation or instruction following, typically benefit from lower temperatures that produce more deterministic outputs.

The article highlights tasks like instruction following, which are essential to many real-world LLM applications. When the goal is to ensure accuracy, adherence to instructions, or stable task performance, lower temperatures are generally preferred. However, for tasks that prioritize novelty, exploration, or unique ideas, increasing the temperature can be advantageous.

One important limitation noted in the paper is that it does not examine coding tasks, even though coding is one of the most common uses of LLMs. While coding often requires deterministic outcomes, there is still flexibility in how a solution can be written. Because both structure and creative variability play a role in programming, future work is needed to explore optimal temperature settings for coding tasks.

Overall, experimenting with temperature can be useful, but it is important to review the output before automating or deploying any changes. Begin with a clear idea of the desired result so that you can evaluate how different temperatures affect it. Some tasks may show little to no change across temperature settings, while others may vary significantly. Testing these hyperparameters yourself can offer practical insight into what works best for your specific needs.

While adjusting temperature can improve LLM performance across many tasks, it does not guarantee perfect results. When using LLMs for sensitive tasks such as translating medical, legal, or safety-critical information, it remains essential to verify accuracy and ensure correctness through human oversight.

References

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